

```
#include <stdio.h>
```

```
#define MAX 50
```

```
#define TIME_QUANTUM 10
```

```
typedef struct
```

```
{
```

```
    int pid;
```

```
    int at;
```

```
    int bt;
```

```
    int rt;
```

```
    int weight;
```

```
    int completed;
```

```
} Process;
```

```
void proportional_share(Process p[], int n)
```

```
{
```

```
    int total_weight = 0;
```

```
    int t = 0;
```

```
    int completed = 0;
```

```
    int gantt[MAX];
```

```
    int g_index = 0;
```

```
    for (int i = 0; i < n; i++)
```

```
    {
```

```
        p[i].rt = p[i].bt;
```

```
        p[i].completed = 0;
```

```
        total_weight += p[i].weight;
```

```
    }
```

```
    while (completed < n)
```

```

{

int ran = 0;
for (int i = 0; i < n; i++)
{
    if (p[i].completed == 0 && p[i].at <= t)
    {
        int time_slice = (p[i].weight * TIME_QUANTUM) / total_weight;
        if (time_slice <= 0)
            time_slice = 1;
        int run_time = (p[i].rt < time_slice) ? p[i].rt : time_slice;
        for (int k = 0; k < run_time; k++)
            gantt[g_index++] = p[i].pid;
        p[i].rt -= run_time;
        t += run_time;
        ran = 1;
        if (p[i].rt <= 0)
        {
            p[i].completed = 1;
            total_weight -= p[i].weight;
            completed++;
        }
    }
}
if (!ran)
{
    gantt[g_index++] = -1;
    t++;
}

```

```

    }
}

printf("\nGantt Chart:\n");
for (int i = 0; i < g_index; i++)
    printf("-----");
printf("-\n");
for (int i = 0; i < g_index; i++)
{
    if (gantt[i] == -1)
        printf(" | ID ");
    else
        printf(" | P%d ", gantt[i]);
}
printf(" |\n");
for (int i = 0; i < g_index; i++)
    printf("-----");
printf("-\n");
printf("0");
for (int i = 1; i <= g_index; i++)
    printf("%5d", i);
printf("\n");
}

```

```

int main()
{
    int n;

    Process p[MAX];

    printf("Enter number of processes: ");

```

```
scanf("%d", &n);

for (int i = 0; i < n; i++)
{
    p[i].pid = i + 1;

    printf("\nProcess P%d\n", i + 1);

    printf("Arrival Time: ");
    scanf("%d", &p[i].at);

    printf("Burst Time: ");
    scanf("%d", &p[i].bt);

    printf("Weight: ");
    scanf("%d", &p[i].weight);
}

proportional_share(p, n);

return 0;
}
```

OUTPUT

Enter number of processes: 3

Process P1

Arrival Time: 0

Burst Time: 10

Weight: 1

Process P2

Arrival Time: 0

Burst Time: 10

Weight: 2

Process P3

Arrival Time: 0

Burst Time: 10

Weight: 3

Gantt Chart:

